Examiner reports

Q2.

Many candidates scored around 10 marks on this question but only the highest achievers scored anywhere near the full 17 marks. In part (a)(i), almost all candidates realised the need for standardisation and so obtained P(Z < 0.909). Far too many candidates then lost the final mark by quoting, from Table 3, P(Z < 0.9) or P(Z < 0.99). In part (a)(ii), most candidates recognised the need for an area change followed by a subtraction of areas but again lost the final mark by compounding an aforementioned error made in part (a)(i). In answering part (a)(iii), at least 50% of candidates undertook a calculation to give an answer other than zero. In part (b), many candidates scored 3 marks by showing a correct method but using z = +1.2816 instead of z = −1.2816. Perhaps a simple sketch would have shown that the value must be less than 69.5.

Answers to part (c) showed a small but welcome improvement. Whilst many candidates considered $\frac{69.25-69.5}{0.55}$ and so scored 0 marks, better candidates did attempt to find the variance or standard error of $\overline{X}$ before standardising. Sadly, many such candidates then failed to find the correct area and so lost 2 marks. Again, a simple sketch would have shown that the answer was greater than 0.5.

Q3.

This was another routine question that was poorly answered by a significant number of candidates. Often these candidates made no real attempt at any part of the question. For well-prepared candidates, this was a straightforward question, testing techniques that feature on the comparable papers in the legacy specifications and, as a result, they scored well.

Most candidates making an attempt at part (a)(i) were successful. However, a surprising number of candidates failed to answer part (a)(ii) correctly; standardising 99 or even 106 was quite common and then these and some other candidates failed to conduct the necessary area change. Most candidates used the tables provided rather than any normal probability facility on their calculators.

In part (b), the success rate by some candidates was high, but again there was a tendency for candidates to use 106 rather than 100 in their standardisation. For those making progress, the usual other error was an incorrect z-value. Some candidates did not show a clear method and they should be advised to show a standardisation equated to a z-value, or equivalent, before attempting any rearrangement.